Trainings

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

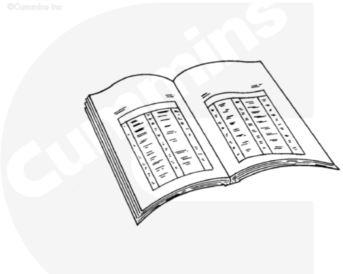
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠


ck800wa

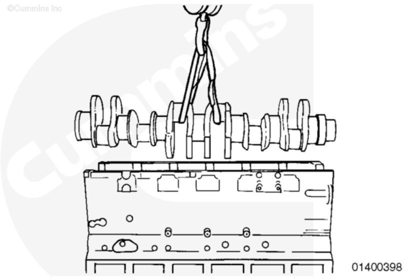
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Remove the engine and place on an engine stand. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/89/89-000-001.html)
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/89/89-016-006.html)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html)
- Remove the connecting rod bearing caps. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
- Remove the main bearing caps. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
- Remove the thrust bearings. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-007.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



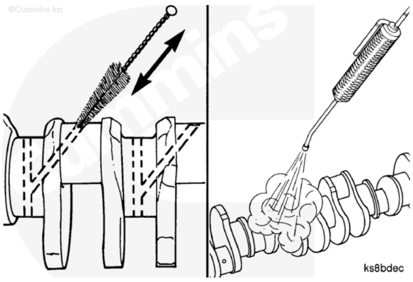
Support the weight of the crankshaft with a hoist or lifting device.

Remove the crankshaft.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



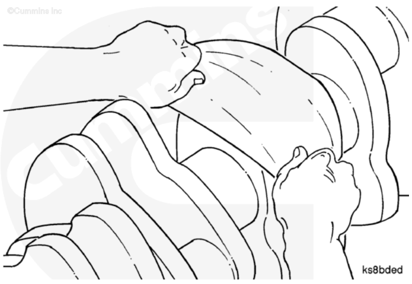
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the crankshaft with solvent and dry with compressed air.

Inspect the crankshaft for nicks or scratches. If a mark can **not** be felt with the fingernail, there is no value in polishing the crankshaft.

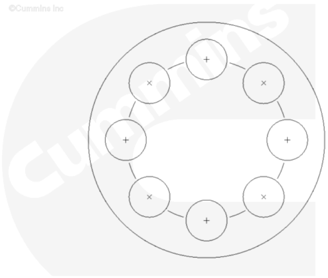
Use a Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, to remove discoloration or light scratches from the machined surfaces.



The crankshaft **must** be cleaned again if this polishing is performed.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a thread chaser to clean the capscrew threads in the crankshaft. Severe engine damage can result.

The QSK23 uses rolled capscrew threads in the bolt holes in the nose of the crankshaft.

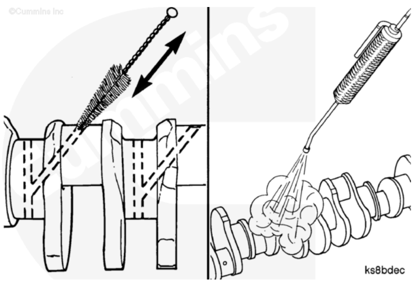
To clean the ROLLED threads, flush with solvent, and dry with compressed air.

If additional cleaning is required, brush with a nylon bristle brush.

Place tape over the threaded capscrew holes.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



ks8bdec

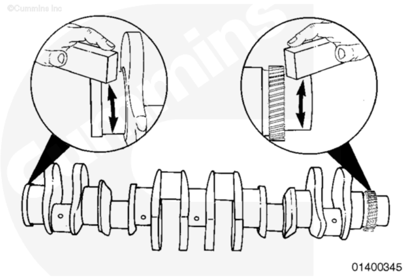
Use a bristle brush and solvent to clean all of the crankshaft oil drillings.

Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

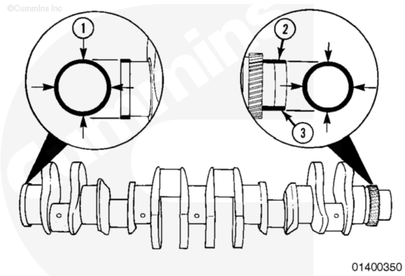
Note : New crankshafts are coated with a heavy preservative. Use solvent to thoroughly remove the coating. Brush or flush the packing debris from the oil drillings before installing the crankshaft into the engine.



Use a honing stone to polish the outside diameter of the front and rear oil seal locations, the flywheel mounting location, and the vibration damper location. Remove all small scratches and grooves.

Note : Use a light preservative oil to prevent rust during engine rebuild. If the crankshaft is not going to be installed immediately, use a heavy preservative oil.

Measure the outside diameter at the locations shown

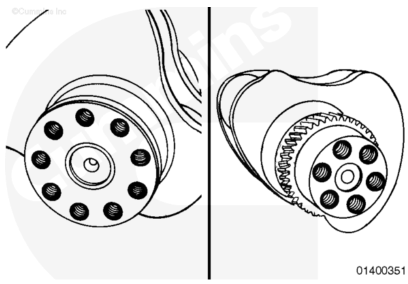


Crankshaft Outside Diameter			
	mm		in
Location (1)	111.075	MIN	4.3730
	111.125	MAX	4.3750
Location (2)	170.481	MIN	6.7119
	170.519	MAX	6.7133
Location (3)	169.981	MIN	6.6922
	170.019	MAX	6.6937

⚠ CAUTION ⚠

Do not chase threads on the crankshaft. Severe engine damage can occur.

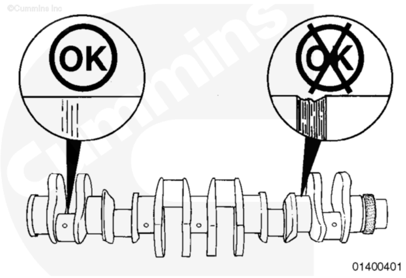
Check the threads for damage at both ends of the crankshaft.



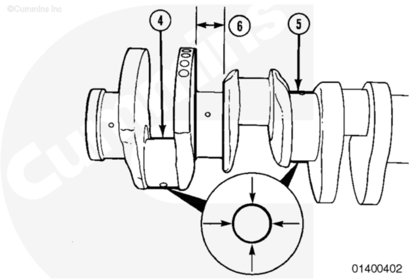
Check the main bearing journals and rod bearing journals for damage or excessive wear. Minor scratches are acceptable.

If a fingernail is caught in the scratch, the crankshaft **must** be replaced or machined.

If the main or rod journals are machined, mark the number 1 counterweight with the amount of material removed. If the crankshaft thrust flanges are machined, mark the adjacent counterweight with the amount of material removed.



Measure the outside diameter.



Rod Bearing Journal Outside Diameter (4)			
	mm		**in**
Standard	107.910	MIN	4.2484
	108.000	MAX	4.2520
Undersize 0.25 mm [0.010 in]	107.660	MIN	4.2384
	107.750	MAX	4.2420
Undersize 0.50 mm [0.020 in]	107.410	MIN	4.2284
	107.500	MAX	4.2320
Undersize 0.75 mm [0.030 in]	107.160	MIN	4.2184
	107.250	MAX	4.2220
Undersize 1.00 mm [0.040 in]	106.910	MIN	4.2084
	107.000	MAX	4.2120

Main Bearing Journal Outside Diameter (5)			
	mm		**in**
Standard	139.910	MIN	5.5083
	140.000	MAX	5.5118
Undersize 0.25 mm [0.010 in]	139.660	MIN	5.4984
	139.750	MAX	5.5020
Undersize 0.50 mm [0.020 in]	139.410	MIN	5.4886
	139.500	MAX	5.4921
Undersize 0.75 mm [0.030 in]	139.160	MIN	5.4787
	139.250	MAX	5.4823

Main Bearing Journal Outside Diameter (5)			
Undersize 1.00 mm [0.040 in]	138.910	MIN	5.4689
	139.000	MAX	5.4724

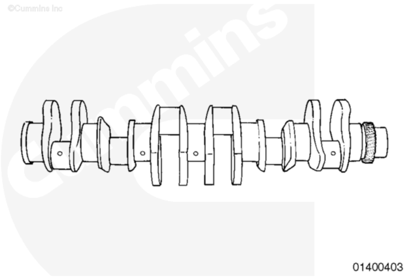
Measure the thrust distance between the thrust faces on the number 6 main bearing journal.

Thrust Distance (6)			
	mm		**in**
Standard	64.000	MIN	2.5197
	64.050	MAX	2.5217
Undersize 0.25 mm [0.010 in]	64.250	MIN	2.5297
	64.300	MAX	2.5317
Undersize 0.50 mm [0.020 in]	64.500	MIN	2.5397
	64.550	MAX	2.5417
Undersize 0.75 mm [0.030 in]	64.750	MIN	2.5497
	64.800	MAX	2.5517
Undersize 1.00 mm [0.040 in]	65.000	MIN	2.5597
	65.050	MAX	2.5617

The crankshaft can be machined undersize if the outside diameter is **not** within specifications. If one of the main journals is ground undersize, **always** grind all of the main journals when one is **not** within specifications. If one of the rod journals is ground undersize **always** grind all of the rod journals when one is **not** within specifications.

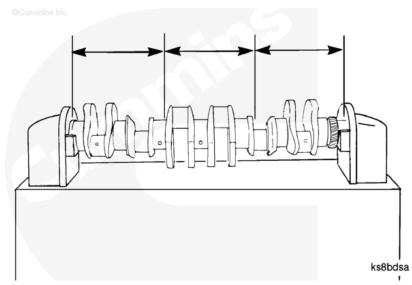
Oversize main, rod, and thrust bearings are available if the crankshaft is machined undersize. Bearings that are 0.25 mm [0.010 in], 0.50 mm [0.020 in], 0.75 mm [0.030 in], and 1.00 mm [0.040 in] oversize are available.

Use a light preservative oil. Lubricate the crankshaft to prevent rust. If the crankshaft is **not** going to be used immediately, use a heavy preservative oil.



Magnetic Crack Inspect

Use a magnetic particle testing machine.



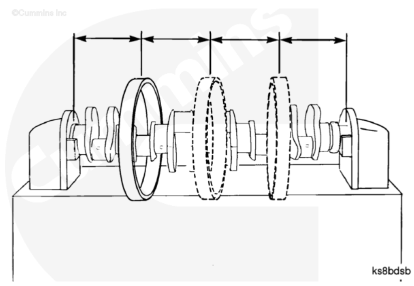
Perform the head shot and inspection method, then perform the coil shot and inspection method.

Adjust the testing machine to 1800 ampere direct current or rectified alternating current.

Use the continuous method. Wet **only** 1/3 of the crankshaft at a time.

Check the crankshaft for cracks.

Use the coil shot method. Use a coil that is a minimum of 514 mm [20.250 in] diameter.



Use the continuous method. Apply coil shot.

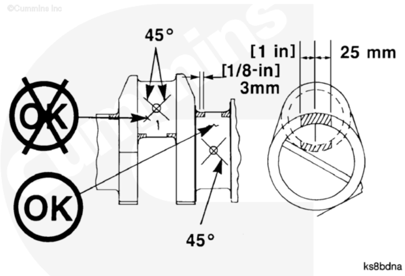
Amperage (Ampere Turns)	
Minimum	**Maximum**
4500 direct current or rectified alternating current	5000 direct current or rectified alternating current

Ampere turns is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check the crankshaft for cracks.

An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface can be seen with the use of the ultraviolet light.

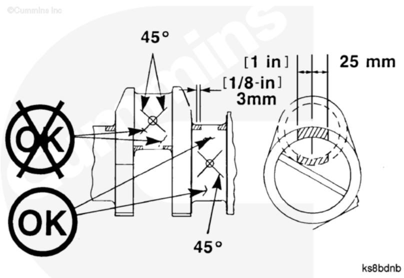
Limits of Acceptance - Open Indications



Do **not** use the crankshaft if:

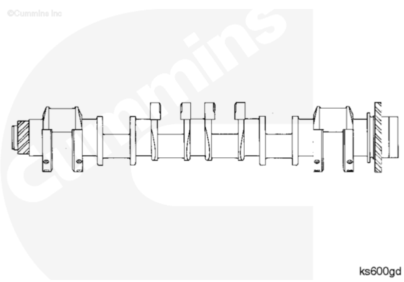
- There is an indication in the fillet or in the shaded area.
- There is an indication that passes through the 45 degree diagonal from the oil hole or goes into the oil hole chamfer.
- There is an indication that is longer than 6 mm [0.250 in].
- There are more than four indications on one journal.

Do **not** use the crankshaft if:



Limits of Acceptance - Indications Below the Surface

- There is an indication in the fillet or in the shaded area that is in a circumferential direction.
- There is an indication in a circumferential direction that is longer than 25.4 mm [1.0 in].
- There is an indication in an axial direction that is longer than 9.5 mm [0.375 in].
- There is an indication that is closer than 1.5 mm [0.063 in] to an oil hole chamfer.
- There is an indication that passes through the 45 degree diagonal from the oil hole.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

The crankshaft must be demagnetized completely and cleaned thoroughly. Small metal particles will cause engine damage.

Use steam to clean the crankshaft and oil drillings.

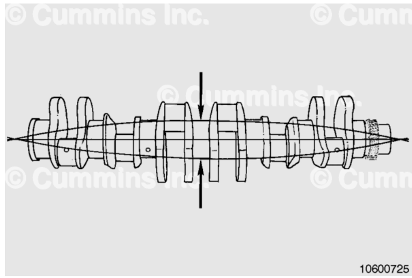
Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.

Use a heavy preservative oil if the crankshaft is **not** going to be installed immediately. Protect the part with a cover to prevent dirt from sticking to the oil.

Bend and Twist Inspect

The crankshaft straightness is determined by the amount of total runout and the adjacent journal runout.

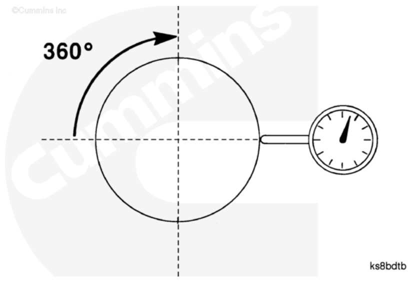
Total runout is defined as the total indicator runout measured at the middle bearing journal, when the crankshaft is supported on the two end journals. Total runout is often referred to as the bend or full length alignment.



Journal runout is defined as the total indicator runout (total sweep of the needle) of the main bearing journals, when the crankshaft is rotated one complete revolution (360 degrees).

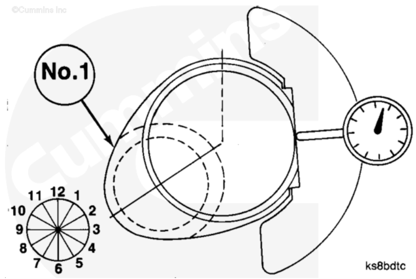
Adjacent journal runout is defined as the relationship of the total indicator runout of a main bearing journal, as it is rotated on a common axis, to the total indicator runout of an adjacent journal.

Adjacent runout is often referred to as step runout, bearing-to-bearing runout, or journal-to-journal runout.



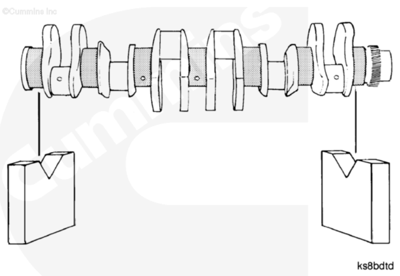
The clock position is defined as the location of the journal at the highest total indicator runout point. Compare its angular relationship with the number one crankshaft pin as viewed from the front of the crankshaft.

In the illustration, the crankshaft pin is at the 8 o'clock position. This is the clock position of the journal being measured.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

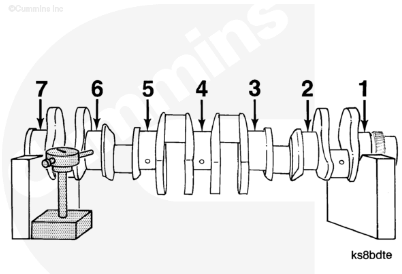


Place two v-blocks on a flat surface.

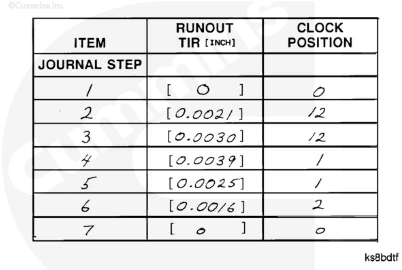
Support the crankshaft on the v-blocks at the two end bearing journals.

Position a dial indicator so the stem touches the center line of the main bearing journal.

The dial indicator **must** be positioned at the centerline of any journal that is measured (1) through (7).



Rotate the crankshaft and measure the total indicator run out at each bearing journal. Record the value and the clock position for each location.



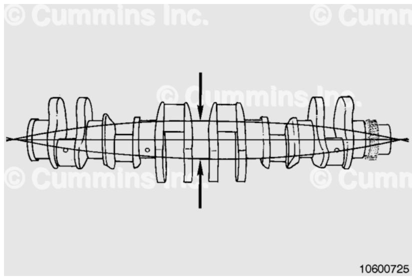
ITEM	RUNOUT TIR [INCH]	CLOCK POSITION
JOURNAL STEP		
1	[0]	0
2	[0.0021]	12
3	[0.0030]	12
4	[0.0039]	1
5	[0.0025]	1
6	[0.0016]	2
7	[0]	0

ks8bdtf

A fully fillet hardened crankshaft **must** be discarded if the total runout is **not** within specifications.

Position the dial indicator stem to the center main bearing journal.

Rotate the crankshaft a complete revolution.



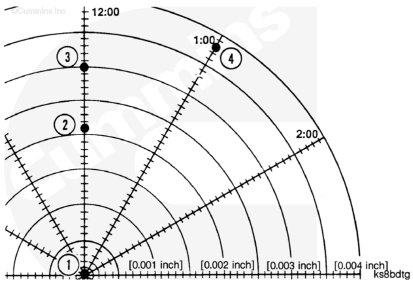
Crankshaft Total Runout		
mm		in
0.150	MAX	0.006

If the crankshaft is **not** within specifications. The crankshaft **must** be replaced.

For each journal, plot the total indicator runout value at its clock position illustrated on the polar chart.

The end journals, supported by v-blocks, **must** be plotted at the center of the chart.

The graphic illustrates the plot points.



Journal	**Total Indicator Run Out**	**Clock Position**
(1)	0	0
(2)	0.002	12
(3)	0.003	12
(4)	0.004	1

Draw a straight line between the plotted points. From journal number one to journal number two from journal number two to journal number three until all journals are plotted on the chart.

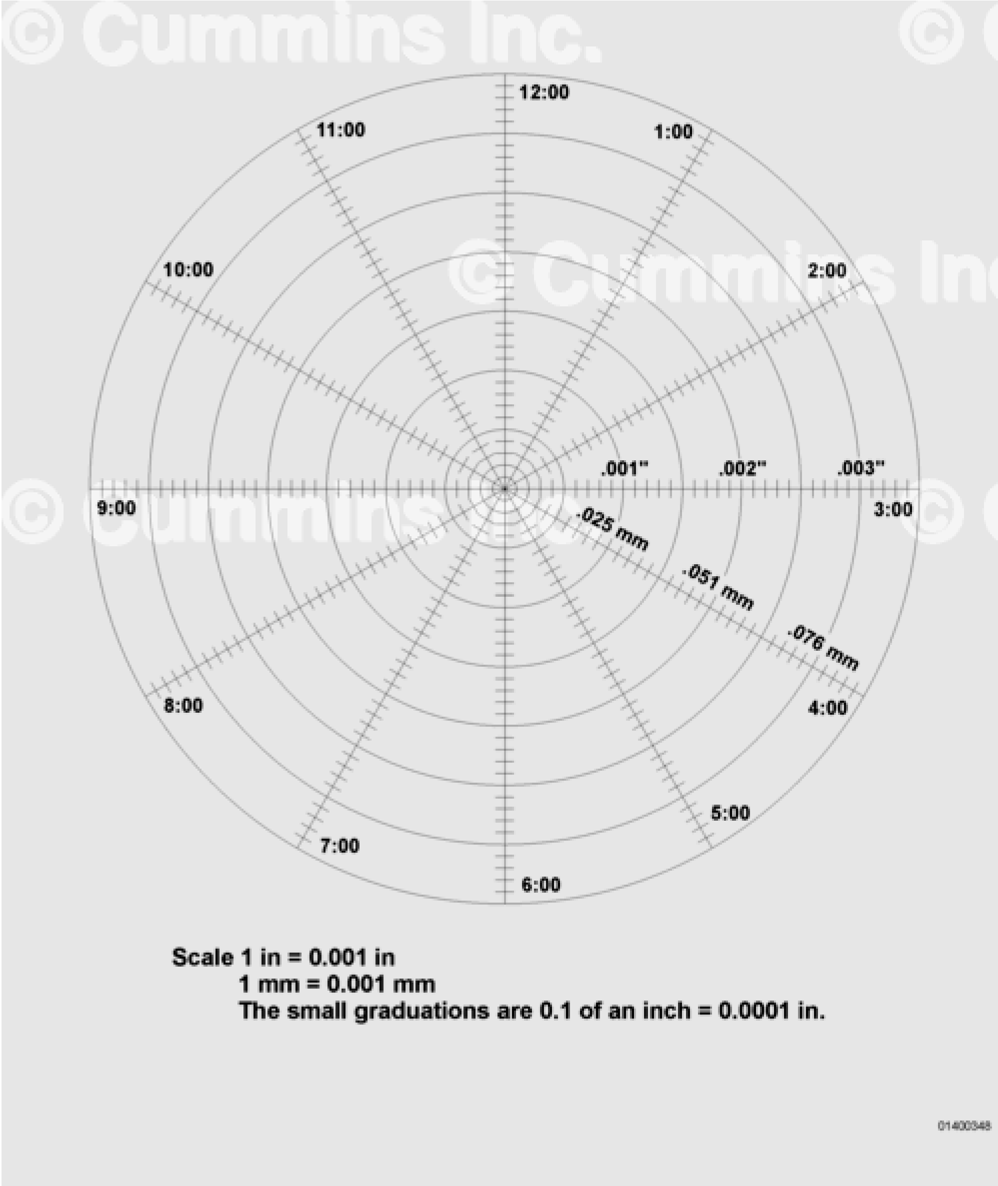
To determine the adjacent journal runout, measure the length of the line from each journal to its corresponding journal point.

In the above table journal number three and number four are 5.1 mm [2 in]. This represents a runout of 0.051 mm [0.002 in].

Record the adjacent journal runout for each main bearing journal.

The maximum adjacent journal runout is 0.070 mm [0.0028 in].

If the crankshaft is **not** within specifications, the crankshaft **must** be replaced.

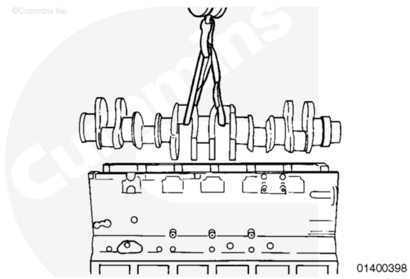


Polar Chart

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Use a lifting strap that will not damage the crankshaft. Do not drop the crankshaft on the bearings.

Use a lint-free cloth to clean the crankshaft bearing journals.

The end of the crankshaft with the smallest diameter **must** point toward the front of the block. Install the crankshaft.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Install the thrust bearings. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-007.html)

- Install the main bearing caps. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
- Install the connecting rod bearing caps. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/89/89-016-006.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)
- Install the engine. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/89/89-000-001.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/89/89-008-018.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

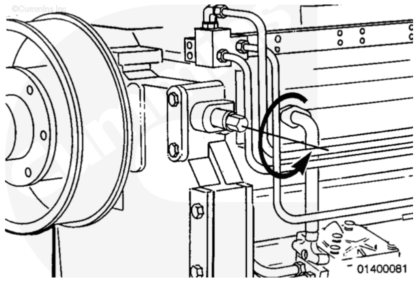
Start the engine and check for leaks.

Rotation Check

To rotate the engine crankshaft, push in on the engine barring device and rotate **counterclockwise**.

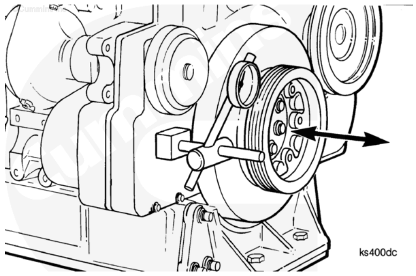
Rotate the crankshaft through two complete revolutions.

If the engine does **not** turn freely, the equipment can have a malfunction. Refer to the equipment manufacturer's instructions. The engine can have internal problems. Refer to the relevant procedure for inspection and replacement of internal engine components.



⚠ CAUTION ⚠

Extreme care must be used in prying against the viscous vibration damper. Sharp pry bars can damage the damper casing, resulting in a leak of the viscous vibration damper fluid and ultimate failure of the vibration damper.



Measure the crankshaft end clearance with a dial indicator.

Measure the end clearance. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)

The check can be made by attaching a dial indicator resting against the vibration damper or pulley while prying against the front cover and inner part of the pulley or vibration damper. End clearance **must** be in specification with the engine mounted in the unit and assembled to the transmission or converter.

If the clearance is **not** within specifications, oversize bearings or machining the crankshaft thrust flanges may be required..

Last Modified: 19-Dec-2011